

Himanshu Kumar

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Education

IIIT Delhi *2022 – 2026 (Expected)*
B.Tech in Computer Science and Engineering CGPA: 6.33/10 (Till 6th Semester)

RSBV Patparganj, Delhi *2019 – 2020*
Class XII, CBSE 84%

RSBV School Block, Shakarpur, Delhi *2017 – 2018*
Class X, CBSE 82%

Relevant Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Algorithm Design & Analysis, Mobile Computing, Machine Learning, Software Engineering, Computer Networks

Technical Skills

Programming Languages:	Python, Java, Kotlin, C, C++, JavaScript, SQL, HTML/CSS
Mobile Development:	Android Studio, Jetpack Compose, CameraX, Android Navigation, Material Design
ML & AI:	PyTorch, PyTorch Mobile, TensorFlow, ResNet-50, CNN, BiLSTM, XLM-RoBERTa, OpenAI SDK, Llama-2
Data & Backend:	MySQL, SQLite, Firebase, REST APIs, Room Database, Scikit-learn, XGBoost, Pandas, Streamlit
Tools:	Git, GitHub, Linux, Docker, VS Code, IntelliJ IDEA, Android Studio, JavaFX
Core Concepts:	Data Structures & Algorithms, OOP, System Design, MVC, Design Patterns, Agile/Scrum

Projects

Plant Disease Detection Mobile Application
Guide: Prof. Arani Bhattacharya | Mobile Computing *Individual | Jan 2024 – May 2025*
- Developed Android app identifying 38+ plant diseases using ResNet-50 with 96% accuracy
- Optimized PyTorch Mobile for offline inference, enabling use in low-connectivity rural areas
- Integrated Llama-2/phi-2 based advisory system for contextual treatment recommendations
- Built responsive UI with Jetpack Compose and live CameraX pipeline

Tech Stack: Kotlin, PyTorch Mobile, Jetpack Compose, CameraX, OpenAI SDK
GitHub Repository

Gendered Abuse Detection in Indic Languages
NLP / Machine Learning Project *Team of 3*
- Built ML models to detect gender-based abuse in Hindi, Tamil, and Indian English social media content
- Implemented CNN-BiLSTM-Attention architecture with XLM-RoBERTa embeddings, outperforming IndicBERT baselines
- Designed multi-task learning system for joint prediction of gendered abuse and explicit content
- Applied transfer learning using domain-adapted Twitter models for cross-lingual abuse detection

Tech Stack: Python, PyTorch, XLM-RoBERTa, CNN, BiLSTM, Attention Mechanisms

AgriPredict: ML-Based Crop Yield Prediction
Machine Learning Project *Team of 4*

- Built ML system for crop yield prediction on a multi-country dataset of 28,242 records
- Implemented ensemble models (Random Forest, XGBoost, Bagging Regressor) achieving 98.9% accuracy
- Created interactive Streamlit GUI for model selection, input, and visualization
- Applied one-hot encoding, feature engineering, and k-fold cross-validation for robust evaluation

Tech Stack: Python, Scikit-learn, XGBoost, Streamlit, Pandas, Matplotlib

Interactive Stick Hero Game
Guide: Prof. Raghava Mutharaju | Advanced Programming *Team of 2 | Nov 2023 – Dec 2023*

- Built 2D game with physics simulation and responsive controls using JavaFX
- Applied MVC and OOP principles for modular, scalable architecture

Tech Stack: Java, JavaFX, OOP, MVC
GitHub Repository

Zoo Management System
Guide: Prof. Raghava Mutharaju | Object-Oriented Programming *Individual | Oct 2023 – Nov 2023*

- Implemented modular system using Factory, Strategy, and Observer design patterns
- Developed CLI with input validation, data persistence, and maintainable architecture

Tech Stack: Java, OOP, Design Patterns

Achievements & Recognition

- Achieved 96% accuracy on plant disease classification across 38 classes using ResNet-50
- Outperformed IndicBERT baseline with CNN-BiLSTM-Attention model for abuse detection in Indic languages
- Recognized by IIIT Delhi faculty for mobile computing project innovation
- Delivered 5+ software projects on schedule across individual and team settings
- Actively contributing to open-source repositories on GitHub

Interests & Hobbies

- **Technical:** AI/ML, mobile development, agri-tech, computer vision, NLP
- **Sports:** Table Tennis, Cricket
- **Other:** Geopolitics